AI Chatbot for University Helpdesk

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Software Requirements Specification (SRS)

An intelligent, 24/7 virtual assistant designed to help university students, faculty, and staff quickly access campus information and support.

Project: AI Chatbot for University Helpdesk

Document: Software Requirements Specification

Version: 1.0

Status: Development

Last Updated: August 2026

Table of Contents

- 1. Introduction
- 2. Purpose
- 3. Scope
- 4. Problem Statement
- 5. Proposed Solution
- 6. Objectives
- 7. Stakeholders
- 8. User Roles
- 9. Functional Requirements
- 10. Non-Functional Requirements
- 11. System Features
- 12. Use Cases
- 13. System Workflow
- 14. System Architecture
- 15. Data Requirements
- 16. AI Requirements
- 17. Security Requirements

- 18. Error Handling
- 19. Performance Requirements
- 20. Testing Requirements
- 21. Technology Requirements
- 22. Assumptions and Constraints
- 23. Future Enhancements
- 24. Acceptance Criteria
- 25. Conclusion

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1. Introduction

Universities handle a large number of repetitive queries related to admissions, courses, examinations, fees, scholarships, hostel facilities, transportation, library services, academic calendars, departments, and campus facilities.

Traditional helpdesk systems often require students to search through multiple webpages, documents, notices, or contact different departments to obtain simple information.

The AI Chatbot for University Helpdesk aims to solve this problem by providing a centralized conversational interface through which users can ask questions in natural language and receive relevant answers quickly.

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2. Purpose

The purpose of this system is to develop an AI-powered university helpdesk chatbot that can:

- Answer frequently asked university-related questions.
- Provide information using natural language.
- Reduce repetitive workload on university helpdesk staff.
- Provide assistance 24/7.
- Retrieve relevant information from university-provided knowledge sources.
- Maintain a simple and user-friendly conversational interface.
- Escalate unanswered or complex queries when required.

The system should act as a first-level digital support assistant for the university.

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3. Scope

The system will provide conversational assistance for common university-related queries.

In Scope

The chatbot may provide information related to:

- Admissions
- Courses and departments
- Examinations
- Fees
- Scholarships
- Hostel facilities
- Library services
- Transportation
- Academic calendar
- University notices
- Campus facilities
- Faculty and department information
- Contact information
- Office timings
- Frequently Asked Questions

Out of Scope

The initial version will not:

- Make final admission decisions.
- Modify official university records.
- Process financial transactions.
- Replace university administrators.
- Provide confidential student information without authorization.
- Make decisions requiring official human approval.

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4. Problem Statement

Students frequently spend significant time searching for university information scattered across websites, PDFs, notices, and departmental resources.

At the same time, university helpdesk staff receive repetitive questions that could be answered automatically.

Existing Challenges

Problem	Impact
Repetitive queries	Increased helpdesk workload
Information scattered across sources	Difficult information discovery
Limited helpdesk availability	Students cannot get immediate answers
Manual searching	Time-consuming
Long response times	Poor user experience
Language complexity	Difficult for some users to understand official documents

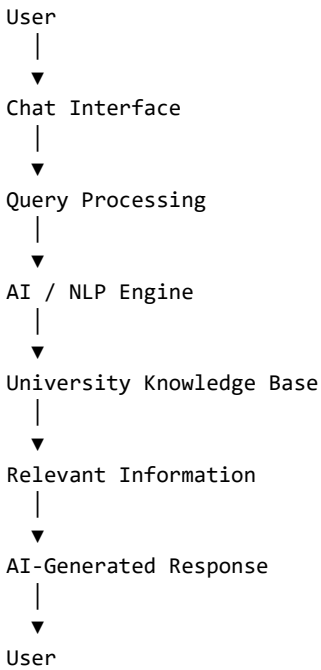
The proposed chatbot addresses these problems through an AI-based conversational interface.

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5. Proposed Solution

The proposed system provides a centralized AI chatbot capable of understanding natural-language questions and generating relevant responses based on authorized university information.

Basic Concept



The system can use university documents, FAQs, notices, databases, or other approved information sources as its knowledge base.

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6. Objectives

The major objectives are:

1. Build an intelligent university helpdesk assistant.
2. Reduce repetitive workload for helpdesk staff.
3. Provide fast and accurate responses.
4. Enable 24/7 information access.
5. Provide a simple conversational user experience.
6. Centralize university information.
7. Reduce the time required to find information.

8. Provide a scalable foundation for future university automation.

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7. Stakeholders

Stakeholder	Responsibility / Interest
Students	Primary users of the chatbot
Faculty	Information and support
Administrative Staff	Manage university information
Helpdesk Staff	Handle escalated queries
University Management	Monitor system effectiveness
Developers	Design and maintain the system
System Administrator	Manage infrastructure and access

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8. User Roles

8.1 Student / General User

Users should be able to:

- Start a conversation.
- Ask university-related questions.
- Receive AI-generated responses.
- Ask follow-up questions.
- View previous conversation history if supported.
- Provide feedback on responses.

8.2 Administrator

Administrators should be able to:

- Manage knowledge sources.
- Add or remove FAQs.
- Update university information.
- Monitor chatbot activity.
- Review unanswered queries.
- Manage system users.
- Monitor system performance.

8.3 Helpdesk Staff

Helpdesk staff may:

- Review escalated queries.
- Respond to unresolved questions.
- Identify frequently asked questions.
- Provide updated information for the knowledge base.

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9. Functional Requirements

FR-01: User Interaction

The system shall provide a conversational interface through which users can submit queries.

FR-02: Natural Language Understanding

The system shall process natural-language questions and identify the user's intent.

FR-03: Answer Generation

The system shall generate responses based on available university information.

FR-04: Knowledge Retrieval

The system shall retrieve relevant information from configured university knowledge sources.

FR-05: Context Awareness

The system should maintain conversational context so that users can ask follow-up questions.

Example:

> *User: When is the next semester exam?*
>
> *Chatbot: The examination begins on 15 December.*
>
> *User: What about the practical exams?*

The chatbot should understand that "practical exams" refers to the same examination context.

FR-06: FAQ Support

The system shall provide predefined answers for frequently asked questions where appropriate.

FR-07: Unknown Query Handling

If the chatbot cannot confidently answer a question, it shall inform the user rather than generating unsupported information.

FR-08: Query Escalation

The system should provide an option to escalate unresolved queries to human helpdesk staff.

FR-09: Feedback

Users should be able to provide feedback about chatbot responses.

FR-10: Conversation History

The system may maintain conversation history for authenticated users.

FR-11: Administrator Knowledge Management

Administrators shall be able to update the chatbot's knowledge sources.

FR-12: Search

The system should support searching relevant university information when required.

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10. Non-Functional Requirements

NFR-01: Performance

The chatbot should respond within an acceptable time under normal system load.

NFR-02: Availability

The system should be available continuously, subject to planned maintenance.

NFR-03: Scalability

The system should support an increasing number of users without major architectural changes.

NFR-04: Security

User information and administrative functionality shall be protected against unauthorized access.

NFR-05: Usability

The chatbot interface should be simple enough for users with minimal technical knowledge.

NFR-06: Reliability

The system should provide consistent responses and handle failures gracefully.

NFR-07: Maintainability

The system should use modular components so that individual components can be updated independently.

NFR-08: Accuracy

Responses should be grounded in trusted university information wherever possible.

NFR-09: Privacy

The system should minimize the collection and storage of unnecessary personal information.

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11. System Features

Intelligent Chat
Users can interact with the chatbot using natural language.

AI-Powered Responses
The system uses AI/NLP capabilities to understand questions and generate useful responses.

University Knowledge Base
The chatbot can use university-specific documents and information as its knowledge source.

Information Retrieval
Relevant information can be retrieved before generating a response.

Contextual Conversations
The chatbot can maintain context across multiple messages.

Human Escalation
Complex or unresolved queries can be transferred to a human helpdesk.

Feedback System
Users can rate or report chatbot responses.

Admin Dashboard
Administrators can manage information and monitor chatbot activity.

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12. Use Cases

UC-01: Ask a Question
Actor: Student

Precondition: Chatbot is available.

Steps:

9. User opens the chatbot.
10. User enters a question.
11. System processes the question.
12. System retrieves relevant information.
13. AI generates a response.
14. Response is displayed to the user.

Postcondition: User receives an answer.

UC-02: Ask a Follow-Up Question

Actor: Student

Steps:

15. User asks an initial question.
16. Chatbot provides an answer.
17. User asks a related follow-up question.
18. System uses previous conversation context.
19. Chatbot provides a contextual response.

UC-03: Unanswered Query

Actor: Student

Steps:

20. User submits a question.
21. System cannot find reliable information.
22. Chatbot informs the user.
23. System provides escalation/contact options.

UC-04: Update Knowledge Base

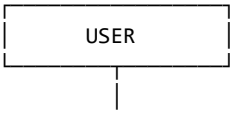
Actor: Administrator

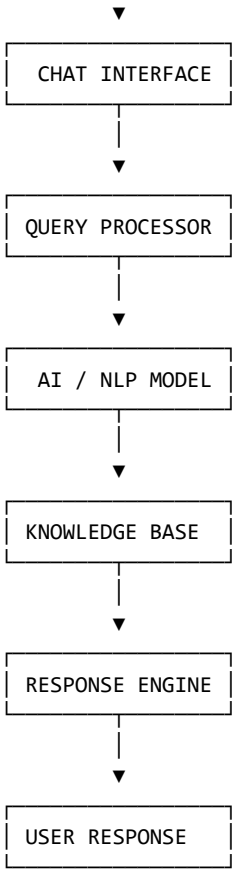
Steps:

24. Administrator logs into the admin panel.
25. Administrator uploads or updates information.
26. System validates the information.
27. Knowledge base is updated.
28. Updated information becomes available to the chatbot.

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13. System Workflow

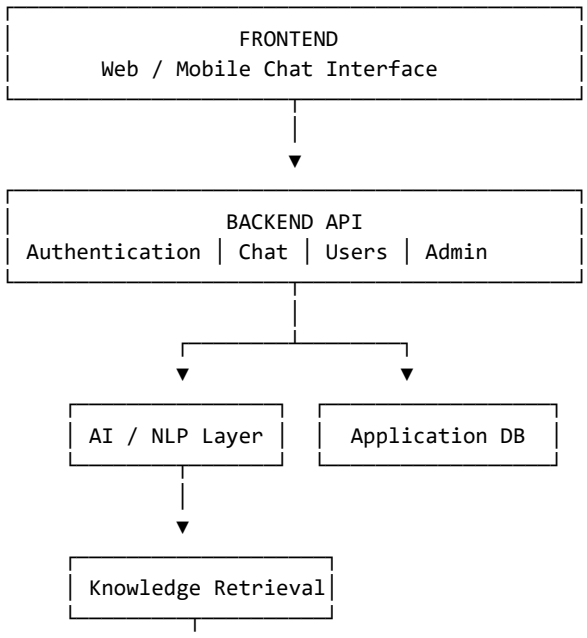


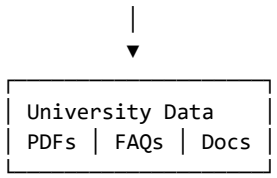


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14. System Architecture

The proposed architecture can follow a modular structure:





If a RAG (Retrieval-Augmented Generation) architecture is used, the AI should retrieve relevant university information first and then generate an answer using that retrieved context.

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15. Data Requirements

The system may manage the following types of data:

User Data

- User ID
- Name
- Email
- Role
- Authentication information

Conversation Data

- Conversation ID
- User ID
- User query
- Chatbot response
- Timestamp
- Feedback

Knowledge Data

- Document ID
- Document title
- Content
- Category
- Upload date
- Last updated date

FAQ Data

- Question
- Answer
- Category
- Status
- Last updated date

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16. AI Requirements

The AI component should:

- Understand natural-language queries.
- Identify user intent.
- Retrieve relevant information.
- Generate understandable responses.
- Maintain conversational context.
- Avoid unsupported claims.
- Handle ambiguous questions.
- Respond appropriately when information is unavailable.

AI Response Principle

The chatbot should prioritize:

Accuracy > Relevance > Clarity > Conciseness

The system should avoid hallucinating university-specific information.

For example:

> ❌ *"The university fee deadline is definitely 15 September."*

when the knowledge base contains no such information.

Instead:

> ✅ *"I couldn't find the current fee deadline in the available university information. Please contact the accounts department for the latest deadline."*

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17. Security Requirements

The system shall implement appropriate security mechanisms.

Authentication

Administrative functionality shall require authentication.

Authorization

Users shall only access features permitted for their role.

Data Protection

Sensitive user information should be protected during storage and transmission.

Input Validation

User inputs shall be validated to prevent malicious or malformed requests.

API Security

Backend APIs should use appropriate authentication and authorization mechanisms.

AI Security

The system should include safeguards against:

- Prompt injection
- Unauthorized information retrieval
- Sensitive information disclosure
- Malicious inputs
- Unauthorized administrative actions

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18. Error Handling

The system should gracefully handle:

- Invalid user input
- Empty queries
- Network failures
- AI service failures
- Database failures
- Missing information
- Unsupported questions
- Authentication failures

Example fallback response:

> *"I'm sorry, I couldn't process that request right now. Please try again or contact the university helpdesk."*

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19. Performance Requirements

The system should be designed to:

- Handle multiple simultaneous users.
- Minimize chatbot response latency.
- Efficiently retrieve relevant documents.
- Avoid unnecessary AI API calls.
- Support caching where appropriate.
- Scale backend services as demand increases.

Performance targets can be finalized after deployment testing.

20. Testing Requirements

The system should undergo multiple levels of testing.

Unit Testing

Test individual functions and modules.

Integration Testing

Verify communication between:

- Frontend and backend
- Backend and database
- Backend and AI service
- AI service and knowledge retrieval system

Functional Testing

Verify that each required feature behaves correctly.

Security Testing

Test authentication, authorization, input validation, and data protection.

Performance Testing

Measure:

- Response time
- Concurrent users
- API latency
- Database performance

AI Evaluation

Evaluate chatbot responses for:

- Accuracy
- Relevance
- Groundedness
- Completeness
- Hallucination rate

21. Technology Requirements

The exact technology stack may vary depending on implementation.

A possible stack is:

Layer	Possible Technology
Frontend	React / HTML / CSS / JavaScript
Backend	Node.js / Express / Python / FastAPI
Database	MongoDB / PostgreSQL / MySQL
AI	LLM API / Open-source LLM
NLP	LangChain / LlamaIndex / Custom Pipeline
Vector Database	FAISS / ChromaDB / Pinecone
Authentication	JWT / OAuth
Deployment	Docker / Cloud Platform
Version Control	Git + GitHub

Note: Replace these technologies with the actual technologies used in the project.

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22. Assumptions and Constraints

Assumptions

- University information is available in a structured or accessible format.
- Administrators regularly update important information.
- Users have internet access.
- The AI service is available when required.
- The knowledge base contains sufficiently accurate information.

Constraints

- AI responses depend on the quality of available information.
- AI-generated responses may occasionally require validation.
- Third-party AI APIs may introduce usage costs or rate limits.
- Internet connectivity is required for cloud-based services.
- University-specific information must be kept up to date.

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23. Future Enhancements

The system can be extended with:

🌐 Multilingual Support

Support languages such as:

- English
- Hindi

- Regional languages

Voice Assistant
Allow users to ask questions using voice.

Mobile Application
Develop dedicated Android/iOS applications.

Notification System
Provide personalized notifications about:

- Exams
- Fees
- Results
- Events
- Important notices

Helpdesk Ticketing
Automatically create support tickets for unresolved queries.

Analytics Dashboard
Display:

- Most frequently asked questions
- Unresolved queries
- User activity
- Response accuracy
- Satisfaction ratings

Personalized Assistance
Authenticated users could receive information relevant to their:

- Department
- Course
- Semester
- Academic status

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24. Acceptance Criteria
The system will be considered successful when:

- [ ] Users can interact with the chatbot.
- [ ] The chatbot understands common university-related questions.
- [ ] Relevant university information can be retrieved.
- [ ] The system generates understandable responses.

- [ ] The chatbot handles unknown questions safely.
- [ ] Users can ask follow-up questions.
- [ ] Administrative users can manage knowledge sources.
- [ ] Basic security mechanisms are implemented.
- [ ] The system handles common failures gracefully.
- [ ] Core functionality passes functional testing.
- [ ] The system can be deployed successfully.

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25. Conclusion

The AI Chatbot for University Helpdesk is designed to transform traditional university support into a more accessible, intelligent, and efficient digital experience.

By combining conversational AI with a trusted university knowledge base, the system can provide students and other users with quick access to important information while reducing repetitive workload for university helpdesk staff.

The architecture is designed to be modular, scalable, secure, and extensible, allowing future capabilities such as multilingual interaction, voice assistance, ticket management, personalized notifications, and advanced analytics to be integrated as the project evolves.

📌 Document Information

Field	Value
Project Name	AI Chatbot for University Helpdesk
Document Type	Software Requirements Specification
Version	1.0
Status	Development
Target Users	Students, Faculty, Staff
Primary Interface	Conversational Chatbot
AI Capability	Natural Language Understanding & Generation
Knowledge Source	University-Approved Information
Repository	GitHub

⭐ Project Vision

"Making university information accessible through a simple conversation."